

NITESH NANKANI

AI Engineer · Voice AI · Conversational Systems · RAG · Agentic Pipelines

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PROFESSIONAL SUMMARY

AI Engineer with 5 production systems deployed across voice AI, RAG pipelines, and multi-agent orchestration. Built India's first Hindi voice AI wellness agent using Twilio, Deepgram, Sarvam AI, and Claude — covering the full STT → LLM → TTS loop with real-world telephony. Deep hands-on experience with Hindi + Hinglish NLP using MuRIL embeddings, code-switching detection, and Indian-language TTS. Backed by 12 years of business operations — I build systems that have to work in production, not just in demos.

CORE SKILLS

Voice AI Stack	Twilio · Deepgram (STT) · Sarvam AI (Hindi TTS) · ElevenLabs · Whisper
LLM & Agents	Claude · GPT-4o · LLaMA 3.1 (Groq) · LangChain · CrewAI · LangGraph · Function Calling
Indian Language NLP	MuRIL embeddings · Hinglish code-switching · Hindi STT/TTS pipelines · BERT fine-tuning
RAG & Memory	ChromaDB · FAISS · Redis caching · Hierarchical RAG · Retrieval pipelines
Backend & Deployment	FastAPI · Docker · Redis · SQLite · WebSockets · REST APIs
ML & Fine-tuning	BERT (7 models fine-tuned) · HuggingFace Transformers · scikit-learn · PyTorch

FLAGSHIP PROJECT — MOST RELEVANT TO THIS ROLE

RecallAI — Hindi Voice AI Wellness Agent

Production-Ready Prototype · Feb 2025 – Present

- Built India's first Hindi voice wellness companion: full STT → LLM → TTS loop over live phone calls.
- Telephony layer: Twilio handles inbound/outbound calls; Deepgram performs real-time Hindi speech-to-text transcription.
- Response generation: Claude LLM processes intent and emotion; Sarvam AI renders natural Hindi TTS output.
- Memory layer: ChromaDB stores conversation history across sessions — agent remembers prior emotional context.
- Emotion detection: fine-tuned BERT model classifies valence and arousal from transcribed speech in real time.
- Designed fallback flows for silence, interruptions, and low-confidence transcription — mirrors production call-centre requirements.
- Architecture: FastAPI backend, async WebSocket pipeline for low-latency voice response.

BargainAI — Hinglish Negotiation Agent

Production Deployed · HuggingFace Spaces · 2025

- Built a Hinglish conversational agent for Indian e-commerce — handles code-switching (Hindi + English mid-sentence).
- Used MuRIL embeddings (Google's multilingual Indic model) for semantic understanding of mixed-language input.
- LLaMA 3.1 via Groq for fast, cost-efficient inference; hierarchical RAG for product-level pricing context.
- Integrated with WhatsApp via Twilio webhook; outbound voice calling via Twilio + Polly.Aditi.
- Demonstrates real-world handling of noisy, informal Indian conversational patterns.

OTHER PRODUCTION PROJECTS

Complaint Intelligence Agent — 4-agent CrewAI pipeline · 7 fine-tuned BERT models · 51,000 Flipkart reviews · 95-99% accuracy · FastAPI + Gradio · Live on HuggingFace

Legal RAG Chatbot — India new criminal codes (BNS/BNSS/BSA/DPDP 2023) · PageIndex + GPT-4o retrieval · BERT classification · Redis + Docker · Live on HuggingFace

AI Job Application Bot — 3-agent CrewAI system: job matcher, resume customiser, cover letter writer · FastAPI backend · Deployed on HuggingFace

WORK EXPERIENCE

Independent AI Engineer (Freelance)

Feb 2025 – Present

AI / ML Engineer · Agent Development · Voice AI · RAG Systems

- Designed and deployed 5 production AI systems spanning voice agents, RAG pipelines, multi-agent orchestration, and Indian-language NLP.
- Built complete voice AI architecture: Twilio + Deepgram + LLM + Sarvam AI — full loop from inbound call to Hindi spoken response.
- Fine-tuned 7 BERT classification models on domain-specific datasets; deployed to HuggingFace Model Hub.
- All projects publicly verifiable: live demos on HuggingFace Spaces, full source on GitHub.

Teespirit · BlinkMart (Founder)

2023 – 2024

Founder & Growth Manager · E-commerce · AI Tools Adoption

- Built and operated a lean print-on-demand e-commerce brand, running all functions solo.
- Early adopter of AI tools in operations: Midjourney (product design), InVideo/Runway (video), ChatGPT (copy + workflow automation), Canva Pro.
- Ran Meta advertising with ROAS and CAC tracking; managed customer acquisition end-to-end.
- Operated Motilone — an AI-generated video content channel.

Kishor & Company

2012 – 2022

Business Head – Sales & Operations · Apparel Manufacturing (KRYSTAL · FOXX · CUTEBOY)

- Led B2B sales, client relationships, and supply chain operations across three apparel brands.
- Managed end-to-end production cycles, vendor negotiations, and multi-city distribution.
- 10 years of operational experience informing how AI systems must perform in real business environments.

EDUCATION

MBA — Marketing	Pune University	Graduated 2012
AI/ML — Self-directed	Edureka Agentic AI · DeepLearning.AI · Anthropic Docs	2025 – Present

WHY THIS ROLE

The iLeads brief asks for someone who can take a voice bot from prototype to production — handling Hindi + Hinglish, noisy audio, interruptions, and real telephony. RecallAI is exactly that system: Twilio + Deepgram + Sarvam AI + Claude, built for real calls, with fallback flows and emotion detection already in the architecture. BargainAI adds the Hinglish code-switching layer using MuRIL — the tool the brief specifically asks for. I haven't handled live call centre volume yet; I'm honest about that. But the architecture is production-ready and I can move it to scale.